A Path Is Not a State

Exact Predictive Refinement of Fixed Connectome Partitions

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Abstract

We ask whether a fixed externally supplied network partition supports exact one-step prediction for a signed nonlinear network model. For finite signed matrices with absolute outgoing-row normalization and unit-gain clipping, sector-mean dynamics factor on eligible unit states if and only if normalized positive outgoing sector profiles are constant within each source sector. These profiles give the coarsest refinement of the fixed partition that determines its original observation's one-step successor on that domain. Applied to 163,439 eligible MaleCNS sources, the four somaSide observation fibers split into 90,148 exact predictive classes; the largest fiber contains 36,896 classes among 72,750 sources. A unique set of three of the four target coordinates separates all classes, although the omitted coordinate certifies an individual failure. The criterion specializes classical block-row-sum logic rather than establishing closure on mixtures or later states. Previously published structural and reached-set audits provide two contrasting tests: complete depth-two and depth-three route lifting coexists with one-step nonfactorization, while a one-step factor on a separate 711-source rational domain lacks a verified image endomap.

Keywords: connectome; coarse-graining; predictive refinement; routed composition; dynamic closure; lumpability.

1 Introduction

A fixed annotation partition gives an economical network description, but may identify sources with different observed one-step successors. We ask how many predictive distinctions remain unresolved inside its observation

fibers, and characterize the coarsest refinement that retains them on eligible unit inputs.

A network summary can also preserve an edge without preserving a path, and preserve a path without preserving a state. Relay compatibility, same-observation successor equality, and successor-image closure are different requirements even when they use the same carrier.

This paper studies them in a model built from the MaleCNS v1.0 connectome [1, 2]. The connectome supplies detected connections and annotations; the signed operator, normalization, clipped update, and observation functions are declared modeling choices.

The primary question is:

> For a fixed network partition, what is the coarsest refinement of eligible unit sources that determines its observed one-step successor?

For the specified signed, normalized, clipped update, positive outgoing sector profiles answer that question exactly. We count their classes in the fixed somaSide and superclass partitions of the retained carrier. The count measures one-step predictions merged by the annotations on eligible unit inputs.

The general characterization makes the descriptive census possible without fitting trajectories. The somaSide annotation merges 163,439 eligible unit sources into four current-observation fibers, but exact profile equality divides them into 90,148 classes of distinct observed one-step successors. This does not merely scale with source-sector size: the slightly larger R fiber has fewer predictive classes than L, while even the 360-source M fiber contains 69. A single exact failure pair is separated by target coordinate M, yet the unique smallest coordinate set that separates *all* somaSide classes is $\{\texttt{<missing>}, L, R\}$. The coordinate needed for one local witness is therefore not necessarily needed for the complete finite predictive refinement.

The supporting comparisons test different operations. On the same carrier, all somaSide routes at depths two and three lift, yet the observed one-step update fails to factor. A separate reached-set study tests whether an exact one-step factor can be iterated. These contrasts locate the predictive-refinement result between routed composition and coarse-state dynamics.

1.1 Relation to the earlier preprints

The structural and reached-set analyses were previously reported in Papers XVI and XVII [3, 4]. New to this manuscript are the unit-source predictive-refinement characterization, the exact full-domain somaSide and coordinate-

resolved census, and a compact rational certificate for a previously identified failure. The earlier results remain separate tests of structural composition and iterative closure. The superclass profile total reproduces Paper XVI's earlier exact grouping result.

1.2 Related work and contribution

Classical lumpability concerns when a projected Markov process retains a closed law [5]; higher-order lumpability studies the effect of retaining observation history [6]. Faccin, Schaub, and Delvenne use entrograms to characterize information flow and memory under network coarse-graining [7]. Our fixed-partition, deterministic unit-source question instead asks for exact equality of observed one-step successors; it neither estimates entropy nor studies an approximate Markov process. Classical strong lumpability tests equality of transition block-row sums [5]. Jacobi and Görnerup give a distinct dual-eigenvector characterization of strong lumpability [8]. Theorem 5.1 has the block-row-sum form for the positive part of the normalized signed matrix because clipping simplifies on unit inputs. It is a model-specific specialization, not a new general lumpability theorem or closure claim for arbitrary activity vectors.

Petrov and Tognazzi study exact and approximate role assignment in multilayer networks, including reduction through backward differential equivalence [9]. Their task constructs role partitions; ours diagnoses distinctions lost by an externally fixed annotation partition, with no approximate role assignment. Equitable partitions support a lower-dimensional epidemic-threshold calculation in a network model [10]; that preservation result does not apply to the partition and update used here. A two-path operator treatment obtains a quotient transfer inequality with an error term and equality criterion [11]. Our relay audit instead asks whether a particular supported two-step candidate has a common microscopic middle node; it supplies no general quotient error bound. These works characterize preserved behavior, construct compatible reductions, or bound quotient errors. Here the partition is fixed externally: we ask how much exact one-step predictive structure remains unresolved inside its fibers.

Positive coarse-graining constructions also have explicit preservation targets. Lu and Vanden-Eijnden [12] construct collective variables preserving region-visit sequence statistics and mean first-passage times. This is not a guarantee that an arbitrary annotation mean is a deterministic closed state. Network-fibration approaches supply maps of dynamical systems under structural compatibility assumptions [13]. Those assumptions cannot be inferred from quotient adjacency alone.

Connectome coarse-graining has been studied through dynamical properties of selected network models [14]. Here the observation is fixed rather than optimized to ensure closure. Theorem 5.1 makes its one-step diagnostic explicit for the clipped signed update, and the complete MaleCNS census quantifies the refinement burden and identifies which target coordinates retain the unresolved distinctions. A compact two-row certificate gives one directly checkable mismatch. Previously published composition and reached-set audits then explain why exact one-step predictive refinement is not by itself a route-composition or iteration theorem.

2 Operation-Specific Criteria

2.1 Paths and routed products

Let $V = \bigsqcup_i V_i$ be a node partition and Q_i its coordinate projectors. For a column-action operator A, a sector word $(i_0, \dots, i_d)$ is quotient-supported when

$$Q_{i_k} A Q_{i_{k-1}} \neq 0 \qquad (1 \leq k \leq d). \tag{1}$$

It lifts when its routed product is nonzero:

$$Q_{i_d} A Q_{i_{d-1}} A \cdots Q_{i_1} A Q_{i_0} \neq 0. \tag{2}$$

Every lifted word is supported: a zero adjacent block would make the product zero. The converse need not hold. Two adjacent blocks can be nonzero while the image of the first lies in the kernel of the second. For nonnegative adjacency, this can mean that the first edge reaches a different relay from the one used by the second. Signed contributions introduce a further possibility of cancellation.

Proposition 2.1 (Nonnegative relay criterion)**.**

Assume A is a finite entrywise nonnegative matrix. For a triple (i, j, k), let

$$\mathcal{I}_{ij} = \{v \in V_j : \exists u \in V_i,\ A_{vu} > 0\}, \qquad \mathcal{O}_{jk} = \{v \in V_j : \exists w \in V_k,\ A_{wv} > 0\}. \tag{3}$$

The triple is quotient-supported if and only if both sets are nonempty. Its routed product is nonzero if and only if $\mathcal{I}_{ij} \cap \mathcal{O}_{jk} \neq \varnothing$. Thus two nonempty disjoint relay sets certify a supported but nonliftable triple.

Proof. Each entry of the routed product is $\sum_{v \in V_j} A_{wv} A_{vu}$. Nonnegativity makes this positive exactly when at least one summand is positive. Such a summand exists for some u, w exactly when a common relay belongs to

both sets. The two adjacent-block conditions require only their separate nonemptiness. □

This criterion is restricted to nonnegative carriers. A shared relay in a signed carrier can still have its contribution canceled by other routes; the signed computations below use their own exact product tests.

The finite liftability proportion is

$$\mathrm{LP}_d = \frac{\#\{\text{nonzero routed sector words of depth } d\}}{\#\{\text{quotient-supported sector words of depth } d\}}, \tag{4}$$

when the denominator is nonzero. It refers to a declared carrier, partition, and depth, not to a probability of biological transmission.

2.2 One observed successor

Let $F : X \to X$ be a deterministic update, $O : X \to Z$ an observation, and $D \subseteq X$ a declared domain. A one-step factorization is a map

$$\overline{F}_D : O(D) \longrightarrow O(F(D)), \qquad O \circ F|_D = \overline{F}_D \circ O|_D. \tag{5}$$

Lemma 2.2 (Fiber criterion)**.**
This factorization exists exactly when

$$O(x) = O(y) \implies O(Fx) = O(Fy) \qquad (x, y \in D). \tag{6}$$

Proof. Necessity follows by applying $\overline{F}_D$ to the common observation. For sufficiency, assign $O(Fx)$ to $O(x)$; the condition makes this independent of the representative. A violating pair rules out exact deterministic factorization on that domain. It need not rule out useful approximate or stochastic prediction. □

For one retained lag, replace the source observation by $(O(Fx), O(x))$ and the target by $O(F^2x)$. Higher history orders must be compared on the same source-time support; otherwise a change in domain can masquerade as an improvement due to memory.

2.3 Iteration is another requirement

One-step factorization does not imply $O(F(D)) \subseteq O(D)$. Without this inclusion the factor map is not an endomap of the verified coarse image. Even if the inclusion holds, agreement with repeated microscopic evolution requires the factorization identity to remain valid on the successor domains. For example, $F(D) \subseteq D$ together with the fiber criterion is sufficient. The finite checks in Section 6 do not supply that condition.

Proposition 2.3 (Verified-image escape)**.**

Suppose the factorization in Lemma 2.2 holds and put $Z_D = O(D)$. If some $x \in D$ satisfies $O(Fx) \notin Z_D$, no endomap $G : Z_D \to Z_D$ agrees with the observed update on every state of D.

Proof. Agreement would require $G(O(x)) = O(Fx)$, but the right side lies outside the codomain of G. □

An escape certificate concerns the declared image. It does not rule out extending the state space and proving a factorization on a larger domain. Conversely, image invariance alone does not establish the factor identity on untested successor states.

2.4 Checks and supporting data

Table 1. Operation-specific obstructions and the data required to check them.

Intended operation	Exact obstruction	Inputs needed to check it
Two-edge composition on a nonnegative carrier	$\mathcal{I}_{ij} \neq \varnothing$, $\mathcal{O}_{jk} \neq \varnothing$, $\mathcal{I}_{ij} \cap \mathcal{O}_{jk} = \varnothing$	Carrier and partition, triple, and complete relay sets or an exhaustive relay scan
One-step factorization on eligible unit states of Theorem 5.1	Same observation sector, but $p_{aj}s_b - p_{bj}s_a \neq 0$ for some j	Two source identities, sector membership, positive sector masses, complete row-normalization sums, and target-sector size
Endomap of a verified observation image	$x \in D$ and $O(Fx) \notin O(D)$	A source-time address, its exact successor observation, and the complete declared image with exact equality checks

These are compact logical certificates, not a claim of minimum encoding size or constant-cost verification. A zero intersection or nonmembership assertion requires exhaustive data for its declared scope. The dynamic certificate uses whole-row normalization sums even when only one output coordinate is needed. Supporting extraction and provenance are therefore part of replay, while the short algebraic witness belongs in the argument. The three types are demonstrated in one connectome case study with explicitly different audit universes: full-release nonnegative coverage, signed-carrier somaSide dynamics, and the reduced superclass reached set.

3 Model, Domains, and Evidence

3.1 Anatomical source and operator

The source is MaleCNS v1.0, with annotation, transmitter-label, and directed body-pair connection-count tables bound in the published provenance record [1–3]. Counts represent detected connections, not measured synaptic efficacy. Transmitter-derived signs are proxies rather than receptor-validated postsynaptic effects [15].

The registered carrier has 211,577 nodes and 26,028,386 directed edges in its full support. Its known-sign subset has 25,214,058 edges. Let W be the signed matrix stored with source rows and target columns, and set

$$s_u = \sum_v |W_{uv}|, \qquad Y_{uv} = \begin{cases} W_{uv}/s_u, & s_u > 0, \\ 0, & s_u = 0. \end{cases} \tag{7}$$

The column-action operator in Section 2 is $A = Y^{\mathsf{T}}$. The source-to-target stored-row route $Q_{i_0} Y Q_{i_1} \cdots Y Q_{i_d}$ is the transpose of $Q_{i_d} A Q_{i_{d-1}} \cdots A Q_{i_0}$, so their zero tests agree. This explicitly connects the route and dynamic conventions without changing the carrier bytes.

3.2 Dynamics and observations

The dynamic closure tests use

$$F_Y(x) = \frac{4}{5}x + \frac{1}{5}C(Y^{\mathsf{T}}x), \qquad C(u)_v = \min\{1, \max\{0, u_v\}\}. \tag{8}$$

External input and stochastic noise are zero. For each nonempty sector,

$$O_i(x) = \frac{1}{|V_i|} \sum_{v \in V_i} x_v. \tag{9}$$

The superclass observation has 28 coordinates; somaSide has four. Missing annotation values are retained as operational labels, not imputed or treated as biological categories.

The negative unit-source tests and the new profile census use all 163,439 eligible sources. The separate reached-set follow-up uses 711 selected unit sources and exact rational orbits through $T = 4$. Static liftability on the signed carrier is exact; the earlier first-reference dynamic scan is numerical. The two-row certificate below proves one exact successor inequality, while the new census uses reduced rational profiles and exact integer counts. Supplementary Information distinguishes the audit domains and gives the diagnostic workflow.

4 Structural Composition Boundary

The previously published nonnegative depth-two audit uses the full edge table and three nested annotation policies [3]. Typed core excludes bodies absent from the curated annotation table; relay strict admits them only at endpoints, and coverage inclusive also admits them as relays. An absent annotation value within the table remains a different, retained label.

Table 2. Supported and lifted two-step sector triples by coverage policy.

Policy	Supported triples	Lifted triples	Liftability
Typed core	6,237	4,970	0.7969
Relay strict	7,019	5,751	0.8193
Coverage inclusive	7,831	6,125	0.7821

The endpoint-only increment adds 782 candidates, of which 781 lift. The relay-inclusive increment adds 812 candidates, but 438 lack a common microscopic relay. Proposition 2.1 identifies the obstruction: nonempty incoming and outgoing relay sets need not intersect. Complete increment counts and the fraction comparison are in Supplementary Information. This nonnegative coverage audit includes bodies outside the signed dynamic carrier; its route counts are not imported into the signed comparison.

On the signed carrier, exact product checks give a separate result:

Table 3. Exact routed-word liftability on the signed carrier at depths two and three.

Partition	Depth	Supported words	Exact nonzero routed words
somaSide	2	64	64
somaSide	3	256	256
superclass	2	4,783	3,667
superclass	3	67,562	41,247

Every registered superclass zero is classified as absence of a microscopic route. All somaSide candidates lift at these two depths; this is the structural half of the same-carrier comparison below.

5 Predictive Refinement of a Fixed Network Partition

5.1 First-order obstruction

The published first-reference scan found different observed successors for unit sources in the same superclass and somaSide sectors [3]. Supplementary Information retains its numerical counts. Theorem 5.1 identifies the exact unit-state criterion, and Section 5.3 supplies a rational somaSide witness. With the signed-route result of Section 4, this gives the same-carrier separation:

$$\mathrm{LP}_2(\mathrm{somaSide}; Y) = \mathrm{LP}_3(\mathrm{somaSide}; Y) = 1, \tag{10}$$

but $O_{\mathrm{somaSide}} \circ F_Y$ does not factor through O_{somaSide} on eligible unit states. The node basis, edge weights, signs, normalization, and partition are fixed across the two tests; products and updates use the transpose-related conventions in Section 3.1. Structural liftability and one-step factorization test different equalities.

5.2 Unit-state factorization by positive sector profiles

The criterion below applies to a finite class of signed network models, independently of the connectome example.

Theorem 5.1 (Unit-state profile characterization)**.**

Let $W \in \mathbb{R}^{N\times N}$ be a real signed matrix on a finite node set partitioned into nonempty sectors $V_1, \dots, V_m$. Put $n_j = |V_j|$, $s_a = \sum_v |W_{av}|$, $E = \{a : s_a > 0\}$, and $D_{\mathrm{unit}} = \{e_a : a \in E\}$. Normalize $Y_{av} = W_{av}/s_a$ on eligible rows and set other rows to zero. Use $O_j(x) = n_j^{-1} \sum_{v\in V_j} x_v$ and the update

$$F_\beta(x) = \beta x + (1-\beta) C(Y^{\mathsf{T}} x), \qquad 0 \le \beta < 1, \tag{11}$$

with $C(u)_v = \min(1, \max(0, u_v))$, unit gain, and zero external input. Define

$$p_{aj} = \sum_{v\in V_j} \max(W_{av}, 0), \qquad r(a) = \left(\frac{p_{a1}}{s_a}, \dots, \frac{p_{am}}{s_a} \right). \tag{12}$$

Then $O \circ F_\beta|_{D_{\mathrm{unit}}}$ factors through $O|_{D_{\mathrm{unit}}}$ if and only if

$$r(a) = r(b) \quad \text{for all } a, b \in E \text{ belonging to the same sector.} \tag{13}$$

When the condition holds, the factor is uniquely determined on $O(D_{\mathrm{unit}})$ by

$$\overline{F}_{\mathrm{unit}}(O(e_a))_j = \frac{\beta\, \mathbf{1}_{\{a\in V_j\}}}{n_j} + \frac{1-\beta}{n_j} r(a)_j. \tag{14}$$

Proof. Since $|W_{av}| \le s_a$, every coordinate of $Y^{\mathsf{T}} e_a$ lies in $[-1, 1]$. Consequently $C(Y^{\mathsf{T}} e_a)_v = \max(W_{av}, 0)/s_a$, giving

$$O_j(F_\beta(e_a)) = \frac{\beta\, \mathbf{1}_{\{a \in V_j\}}}{n_j} + \frac{1-\beta}{n_j} \frac{p_{aj}}{s_a}. \tag{15}$$

Two unit sources have equal observations exactly when they belong to the same sector: their observation has a single positive entry at that sector. For such a pair the leak terms agree, and

$$O_j(F_\beta(e_a)) - O_j(F_\beta(e_b)) = \frac{1-\beta}{n_j}\big(r(a)_j - r(b)_j\big). \tag{16}$$

Every multiplier is positive. Necessity follows from Lemma 2.2, and sufficiency follows by the displayed construction, which is independent of the representative exactly when the profiles agree. Only sectors containing eligible sources contribute to its domain. □

Corollary 5.2 (Compact profile obstruction)**.**

For integer W, let a, b be eligible sources in the same observed sector. The data

$$(a, b, j;\ p_{aj}, p_{bj}, s_a, s_b, n_j) \tag{17}$$

certify failure of unit-state factorization whenever $p_{aj} s_b - p_{bj} s_a \neq 0$. The exact coordinate difference is

$$\frac{(1-\beta)(p_{aj} s_b - p_{bj} s_a)}{n_j s_a s_b}. \tag{18}$$

The sums in this certificate must be established from the declared carrier. In particular, s_a, s_b include the complete absolute outgoing rows; retaining only the two edges into coordinate j would not suffice. This is a compact algebraic witness, without a claim of smallest encoding.

Theorem 5.1 identifies what the observation forgets: it merges source nodes without requiring their normalized positive outgoing sector profiles to agree. The substochastic kernel $K_{av} = \max(W_{av}, 0)/s_a$ on eligible rows has $r(a)$ as its block-row sums. This is the familiar strong-lumpability pattern [5]; the theorem derives its exact role for the specified clipped dynamics on unit inputs.

The domain qualification is essential. With one sector and $W = \left(\begin{smallmatrix} 1 & -1 \\ -1 & 1 \end{smallmatrix}\right)$, both eligible unit sources have profile $(1/2)$, so the theorem gives factorization on those sources. But $x = e_1$ and $y = (e_1 + e_2)/2$ have the same mean, while $O(F_\beta x) - O(F_\beta y) = (1-\beta)/4 > 0$. Thus the unit-state condition does not imply factorization even on all mixtures of those sources. At $\beta = 1$ the update is the identity and profile equality is unnecessary; for gains above one, first-step upper clipping can invalidate the profile formula.

5.3 The somaSide certificate

Use the previously recorded L-sector representative pair: body IDs 10002 and 10006, at zero-based carrier indices 1 and 4. Both have the registered acetylcholine proxy sign +1 and positive outgoing mass. The observation-coordinate order is (<missing>, L, M, R), with sizes (60851, 75215, 392, 75119). Complete extraction of their two carrier rows gives:

Table 4. Exact outgoing-sector data for the two-source somaSide certificate.

Source body	Edges	s_a	$p_{a,\text{missing}}$	$p_{a,L}$	$p_{a,M}$	$p_{a,R}$
10002	139	3369	0	191	1	3177
10006	131	704	12	523	11	158

Their common input observation is $(0, 1/75215, 0, 0)$. The compact Corollary 5.2 certificate is

$$(10002, 10006, M;\ 1, 11, 3369, 704, 392). \tag{19}$$

These numbers denote body IDs, the retained coordinate, and the certified sums; the complete replay also fixes their carrier indices and partition. At coordinate M,

$$p_{a,M}s_b - p_{b,M}s_a = 1 \cdot 704 - 11 \cdot 3369 = -36355, \tag{20}$$

and therefore

$$O_M(F_Y(e_a)) - O_M(F_Y(e_b)) = \frac{1}{6603240} - \frac{1}{125440} = -\frac{661}{84521472} \neq 0. \tag{21}$$

This proves exact first-order nonclosure of the registered rational model on its eligible unit-state domain, alongside exact finite static lifting on the same carrier and partition. Neither the source sector nor the displayed target coordinate is a missing-annotation bucket.

The algebraic obstruction is $r(a)_M = 1/3369 \neq 11/704 = r(b)_M$. The supporting record retains all 270 outgoing edge entries, both row sums, and carrier-file digests. Its arithmetic and source-extraction checks are distinct; Supplementary Information gives their provenance.

5.4 Predictive profile multiplicity within observation sectors

Let $E_i = E \cap V_i$, $N_i = |E_i|$, and define

$$M_i = \#\{r(a) : a \in E_i\}. \tag{22}$$

This is the predictive profile multiplicity within one observed source sector. More precisely, the partition

$$\mathcal{P}_{\mathrm{pred}} = \big\{\{a \in E_i : r(a) = u\} : i = 1, \dots, m,\ u \in r(E_i)\big\} \tag{23}$$

is the coarsest refinement of the fixed annotation partition on E for which $a \mapsto O(F_\beta(e_a))$ is constant on every class. Indeed, Theorem 5.1 makes its classes successor-constant, while any successor-constant refinement must separate two sources in one sector whenever their profiles differ. Thus $\sum_i M_i$ is the exact number of classes required for this restricted one-step prediction task, not the number of states of an autonomous quotient dynamics. By Theorem 5.1,

$$M_i = \#\{O(F_\beta(e_a)) : a \in E_i\}. \tag{24}$$

The theorem's affine profile-to-successor map is injective for fixed i, so $M_i > 1$ counts distinct first-step predictions merged by one current observation.

We compute the profiles of all 163,439 eligible sources on the retained integer-weight carrier. For exact grouping, retain the source sector and encode the rational profile $r(a)$ by its canonical integer key

$$g_a = \gcd(s_a, p_{a1}, \dots, p_{am}), \qquad k_a = (s_a/g_a, p_{a1}/g_a, \dots, p_{am}/g_a). \tag{25}$$

For eligible sources, $g_a > 0$ and $r(a) = r(b)$ if and only if $k_a = k_b$. The key is an encoding, not another profile: when $r(a) = 0$, its key is $(1, 0, \dots, 0)$, not the all-zero vector. No tolerance or trajectory fitting enters this comparison. The exact superclass total is 119,138, reproducing the previously reported grouping total [3]. The new somaSide total is $\sum_i M_i = 90148$:

Table 5. Exact one-step predictive classes within each somaSide source sector.

somaSide source sector	Eligible sources N_i	Exact profiles M_i	Largest profile class
`<missing>`	17,850	14,535	528
L	72,479	38,648	10,868
M	360	69	291
R	72,750	36,896	10,890

R has 271 more eligible sources than L but 1,752 fewer predictive classes; M has only 360 sources but retains 69 classes. Source-fiber size alone therefore does not determine multiplicity. The largest eligible somaSide fiber,

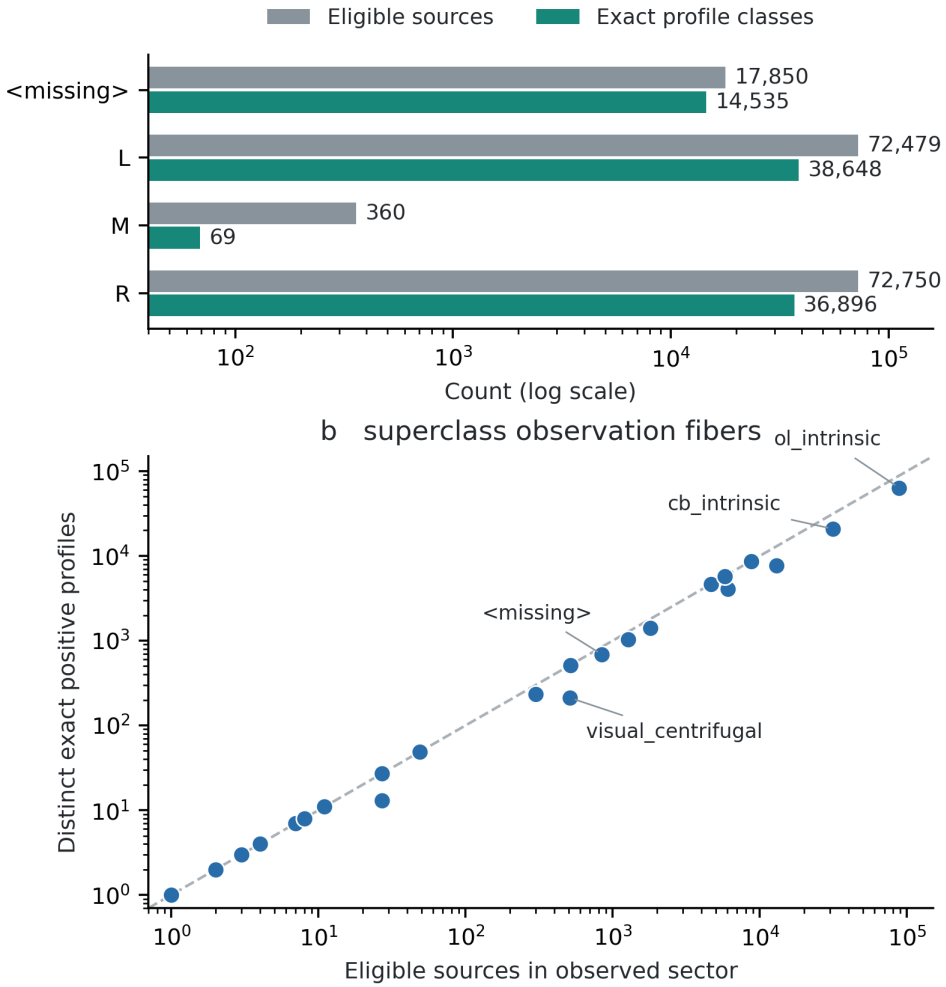


Figure 1. Predictive profile multiplicity. Top: eligible source counts and exact profile classes in each somaSide observation fiber. Bottom: superclass fibers with eligible sources; the dashed line marks one class per source. Axes use logarithmic scales. Counts refer to unit sources in the declared carrier and are not biological state counts.

Alt text: Two-panel chart. The top panel compares eligible unit sources with distinct one-step profiles in the four somaSide sectors: missing 17,850 versus 14,535, L 72,479 versus 38,648, M 360 versus 69, and R 72,750 versus 36,896. The bottom panel plots the corresponding counts for superclass sectors on logarithmic axes against a one-class-per-source diagonal; profile counts do not exceed source counts.

R, contains 36,896 distinct predictions among 72,750 sources. For superclass, the largest eligible fiber is `ol_intrinsic`: 88,519 sources and 63,331 profiles.

To specify which coordinates most frequently split a fiber, let n_{ijq} count sources in E_i with exact coordinate value $r(a)_j = q$. Then

$$D_{ij} = \binom{N_i}{2} - \sum_q \binom{n_{ijq}}{2} \tag{26}$$

is the exact number of unordered same-observation source pairs separated by target coordinate j. We report both the number of source sectors with $D_{ij} > 0$ and the source-weighted sum $\sum_i D_{ij}$. These rankings answer different questions. A pair may be separated by several coordinates, so the counts cannot be added as disjoint failures.

For somaSide, every coordinate splits all four source fibers, but L separates the most within-sector pairs (5,008,763,423) and M the fewest (1,307,422,058), out of a common denominator of 5,432,178,301. Figure 2 compares normalized rates; Supplementary Table S5 gives every exact absolute count. For superclass, `visual_projection` separates the most source pairs, whereas `descending_neuron` splits the most source sectors.

A pairwise mismatch and a coordinate set separating every class must be distinguished. Any given unequal pair needs only one differing coordinate. For the latter task, define

$$M_i(J) = \#\{r(a)|_J : a \in E_i\}. \tag{27}$$

The retained coordinate set J is sufficient exactly when $M_i(J) = M_i$ for every source sector. Exhausting all 16 somaSide coordinate subsets gives the unique minimum set

$$J_{\min} = \{\texttt{<missing>}, L, R\}, \qquad |J_{\min}| = 3. \tag{28}$$

Necessity has short witnesses: source body 30930 has $r(30930) = 0$, whereas bodies 13535, 11139, and 13325 have profiles $(1,0,0,0)$, $(0,1,0,0)$, and $(0,0,0,1)$ respectively. All four sources belong to the retained missing-label source fiber. Omitting the respective nonzero coordinate merges each indicated pair even if all other coordinates are retained. The complete census verifies sufficiency of the three coordinates. This is a minimum for the declared finite dataset including its missing-label bucket, not a universal coordinate reduction law. Coordinate M remains a valid certificate for the pair in Section 5.3 despite being dispensable for this full-data separation task. The local witness asks whether *one pair* differs; the minimum set asks whether *every distinct class* remains separated after projection.

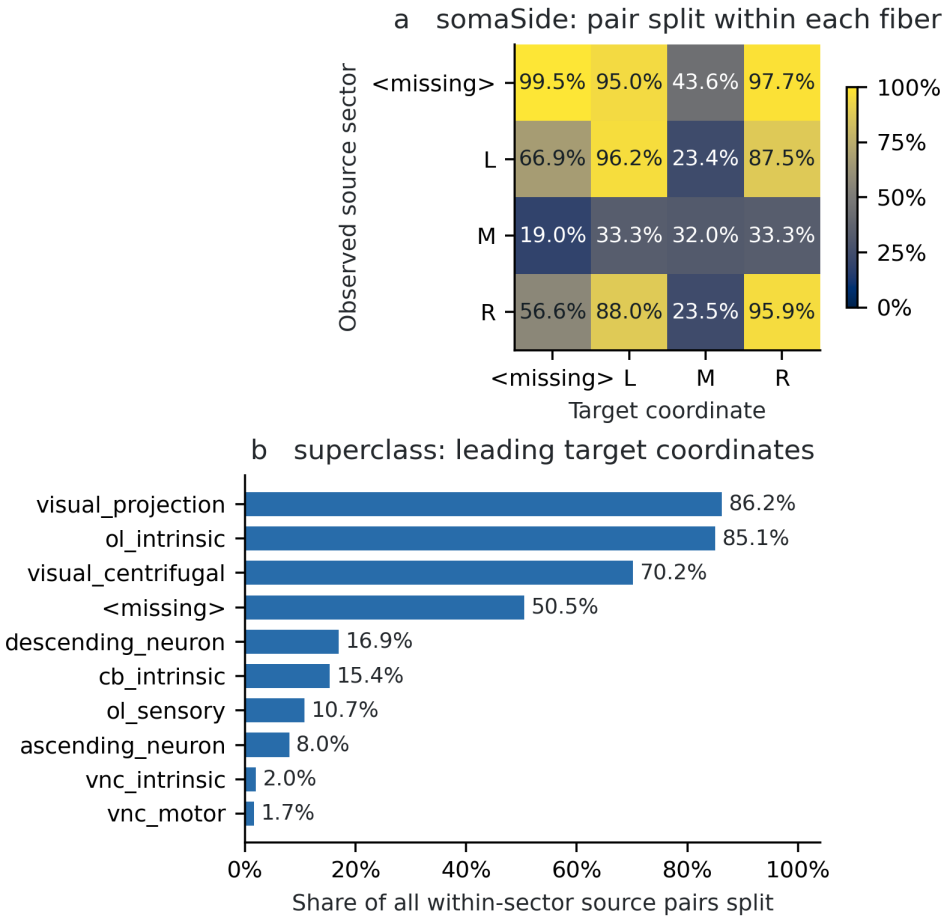


Figure 2. Coordinate-specific profile splitting. Top: $D_{ij}/\binom{N_i}{2}$ for somaSide. Bottom: the ten largest superclass values of $\sum_i D_{ij}/\sum_i \binom{N_i}{2}$. The bottom panel weights sectors by their number of source pairs; the top uses a separate denominator for each row. Displayed percentages are rounded; the underlying counts are exact.

Alt text: Two-panel chart. The top heat map gives the percentage of within-sector source pairs separated by each of four target coordinates for each somaSide source sector; M is least discriminating in the large L and R sectors. The bottom bars rank ten superclass target coordinates by their pooled pair-separation share; visual_projection is highest at 86.2%.

For the 28-coordinate superclass observation, we certify nine individually mandatory coordinates and one sufficient set of 12; the global minimum is not determined by the present analysis. This result is reported as bounds, not as a solved feature-selection problem.

Large classes with $r(a) = 0$ also have an explicit modeling explanation. Every edge in a given source row has the same transmitter-proxy sign; a negative row has no positive first-step drive and hence $r(a) = 0$. The somaSide L, M, and R zero classes contain 10,868, 291, and 10,890 sources respectively. These are model-imposed first-step equivalences; the census does not identify biological redundancy, later-memory equivalence, or a neural mechanism.

6 Reached-Set Factorization and Image Escape

6.1 The domain changes

An earlier superclass one-lag audit found an exact equal-history pair in the named sector `cb_intrinsic` with nine unequal next-step coordinates [3]. The distinct Paper XVII follow-up [4] selects D_0 of 711 unit sources and retains exact rational orbits $x_{a,t} = F_Y^t(e_a)$ through $T = 4$. Its original history-order tests used different starting times. On the common domain $\Omega = D_0 \times \{2, 3\}$, all three history orders $h = 0, 1, 2$ induce the same partition of 1,422 windows into 1,228 exact observed values. The observation alone determines the next observation on that domain; adding two lags separates no further pairs. Supplementary Table S6 records the full history counts and source-selection rule.

6.2 Which fibers remain?

The earlier unsafe source pairs become distinguishable by $t = 2$ on the selected set. The remaining non-singleton fibers are successor-compatible on both audited slices. Appendix A shows that they contain distinct microscopic states but have zero clipped drive, so common leakage explains their one-step equality. The five cohort sizes and redundant-window count are retained in Supplementary Information.

6.3 Why this is not an autonomous coarse state

Each of the $t = 2, 3, 4$ observed slices contains 614 values. The first two are disjoint, forming the verified image $\mathcal{Z}_{2,3}$ of size 1,228. The $t = 2$ successors remain in this image, but all 614 values at $t = 4$ lie outside it. Thus the

factor has one verified internal shift and fails the endomap condition of Proposition 2.3 on $\mathcal{Z}_{2,3}$. This is the required counterexample to inferring iterable coarse dynamics from the one-step factor.

7 Interpretation and Limitations

The census separates two sources of within-fiber equality. Clipping maps eligible negative-sign rows to a zero positive profile, creating model-induced first-step equivalence. Distinct nonzero profiles instead expose positive-routing heterogeneity within the fixed annotation. The contrast between the local M-coordinate witness and the global three-coordinate separating set shows why detecting one failure and reconstructing the entire predictive partition are different tasks.

The structural and reached-set examples test adjacent preservation questions. Relay compatibility is needed for routed composition; an observation fiber needs equal successors for one-step prediction; an iterable coarse state additionally needs a verified image endomap. In the Paper XVII follow-up, zero clipped drive explains the persistent fiber equality under common leakage, while image escape prevents iteration on the audited image.

The partitions are external annotations, the signs are transmitter-based proxies, and the dynamics is a declared model on one connectome release. Theorem 5.1 has a classical block-row-sum form specialized to that model and unit inputs. The exact census describes its predictive distinctions; it does not establish physiological mechanism, forecasting accuracy, or robustness to alternative sign and partition policies.

8 Data and Reproducibility

Published source packages and verification entry points are available for Paper XVI and Paper XVII. The versioned census Dataset [16] contains the exact full-domain profile result, its source replay, and the figure-generation script. A separate compact two-source witness supplement contains the exact certificate and the three Paper XVI metadata records needed for source extraction. Source replay of that supplement additionally consumes the byte-pinned carrier files in the Dataset. Supplementary Information identifies the distinct input domains and reproduction levels. The Dataset is deposited at Zenodo under DOI doi:10.5281/zenodo.23555888.

9 Conclusion

For the signed clipped update on eligible unit sources, positive outgoing sector profiles give the exact one-step predictive refinement of a fixed network partition. In the MaleCNS carrier, four somaSide observation fibers contain 90,148 such classes. The class counts and the unique minimum three-coordinate separating set quantify distinctions lost by the original annotation, rather than proposing another optimized partition.

Structural composition and iteration require different information. The same carrier lifts every somaSide route at depths two and three but fails first-order factorization. On a separate reached-set domain, an exact one-step factor has one internal shift before its successor leaves the verified image. Together, these examples specify what the fixed partition predicts, where that prediction is exact, and which additional operation needs another preservation test.

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Appendix A Noninjectivity and Drive-Off

The equal-observation cohorts in Section 6 are not merely duplicate addresses for equal microscopic states. Write $\beta = 4/5$. The clipped update preserves $[0,1]^V$. For distinct unit sources $a \neq b$,

$$(x_{a,t})_a \geq \beta^t, \qquad (x_{b,t})_a \leq 1 - \beta^t. \tag{29}$$

The first inequality follows by dropping the nonnegative clipped term; the second follows by bounding it by one and iterating from zero. At $t = 2, 3$, $\beta^t > 1/2$. Cohort members therefore remain distinct microscopic states while sharing their sector-mean observations.

The retained effective-drive audit covers all 102 persistent sources at saved current times $t = 1, 2, 3$, giving 306 caches. Every raw drive is nonzero and coordinatewise nonpositive, and every clipped drive is zero. In particular, it covers all 204 members of the ten non-singleton fibers on the common support. Consequently, for these states,

$$F_Y(x) = \frac{4}{5}x, \qquad O(F_Y x) - O(F_Y y) = \frac{4}{5}[O(x) - O(y)] = 0. \tag{30}$$

Their microscopic differences are scaled, not erased. The saved evidence therefore supplies a common-leak explanation of the observed one-step equality on these fibers.

The audit recomputes clipping from saved exact raw-drive payloads. It does not regenerate $Y^{\mathsf{T}} x_t$ or the orbits, classify all 711 sources as drive-off, or prove robustness under arbitrary removal of source blocks. It also does not establish the cause of the initial unsafe-fiber split. These limitations are compatible with exactness of the stated finite factorization.

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